

Neural Network Fuzzy Electricity Demand Forecasts Based on Fuzzy Inputs

Abstract—Predicting electricity demand is crucial for the stability and reliability of the smart grid, especially when incorporating renewable energy sources. Recently, there has been a growing interest in studying both long-term and short-term forecasts of electricity demand using dynamic regression models with seasonal ARIMA (SARIMA) errors and neural network autoregressive (NNAR) models. In the literature, point forecasts of the temperature are used to obtain fuzzy demand forecasts. The superiority of fuzzy forecasts obtained by incorporating randomness and fuzziness in option pricing and fuzzy algo trading has been demonstrated. In electricity demand forecasting, the quality of the temperature input data determines the quality of the demand forecasts. Temperature is a dynamic variable in the sense that it fluctuates throughout the day, and it is important to account for its hour-to-hour and seasonal variation. In this paper, first, several time series models and neural net models are fitted to temperature data, and the data-driven best forecasting model has been selected. Then, the resulting fuzzy forecasts of temperature are used as fuzzy input to obtain the fuzzy demand forecasts. Unlike the existing work, the novelty of the paper is to use fuzzy forecasts of temperature to obtain the fuzzy demand forecasts. The superior performance of neural network fuzzy electricity demand forecasts for Ontario electricity demand data is also discussed in some detail.

Index Terms—Electricity demand, fuzzy forecasts, neural network autoregressive models, dynamic regression models

I. INTRODUCTION

A balanced electricity grid depends on accurate electricity demand forecasts that can fulfill the electricity load requirement of users. For a given problem, the input data of a learning algorithm is almost always expressed by several features (attributes or variables). Recently, both long-term and short-term forecasts have been studied under different approaches, and in some cases, long-term forecasts based on seasonal models may extend to several years. When models incorporate features such as calendar characteristics and meteorological effects (temperature and cloud cover), machine learning models perform better than traditional models such as Autoregressive Integrated Moving Average (ARIMA) model. Both statistical and machine learning techniques have been used extensively to obtain accurate electricity demand forecasts with different features such as temperature, cloud cover, and wind speed [1]. The studies [2], [3] investigate different approaches such as exponential smoothing, regularized forecasts, neural networks, and Kalman filtering in this context. These studies emphasized the efficiency of the methods mentioned above in short-term forecasts with small historical data.

Even though the machine learning models are efficient and provide accurate forecasts, most of them, except neural network autoregressive (NNAR) models and neural network

regression models for time series, are often difficult to interpret due to the model's complexity and lack of transparency. This issue is addressed by combining the recently proposed neural network models with statistical models [1], [2], [4]. For example, adaptive generalized additive models are introduced incorporating Kalman filters, and they can adjust to new electricity consumption patterns in [3].

Real-world problems, especially electricity demand, deal with uncertain, dynamic, and imprecise data. Since the development of fuzzy set theory in 1965 by Zadeh [5], there has been a growing interest in the fuzzy approach in almost all areas. The concept of fuzzy sets has been investigated in the literature to provide clear insights for designing forecasting models. For instance, one key benefit of fuzzy time series approximations is the capacity to operate with minimal data sets without the linearity assumption.

There has been a growing interest in combining randomness and fuzziness to model volatility forecasts, algorithmic trading [6], financial option pricing [7], and to solve portfolio optimization problems. It has been demonstrated in the fuzzy option pricing context that fuzzy forecasts outperform minimum mean square forecasts. However, fuzzy methods remain challenging to apply in practice.

Forecasts of a given time series can be obtained using benchmark models such as the mean (average) model, naive model, seasonal naive model, and drift model. Even though they are simple and easy to implement with less computational burden, their performances are limited in practice as they fail to capture the complicated correlation structures and nonlinearity in real-life data. Thus, to obtain more accurate forecasts, more advanced nonlinear time models such as NNAR have been considered recently. Moreover, forecast performance has been investigated by combining dynamic regression models and neural network regression models [14].

Seasonal ARIMA models can be used to model a wide range of seasonal data [8], [9]. In order to include other relevant information along with information from past data, the dynamic regression models with ARIMA/SARIMA errors are more appropriate and efficient [2]. A recently proposed industrial model to obtain time series forecasts with the given feature values is the Prophet model [10]. Prophet is an additive model which captures the non-linear trend, seasonality, and holiday effects. Even though it is robust to missing data and handles outliers and shifts in the trend, it requires time series that have strong seasonal effects and several seasons of historical data.

Machine learning models are becoming popular in time

series forecasting due to their accuracy and capability to handle more complex real-world nonlinear problems. The feed-forward neural network is a popular way to approximate a given multivariate nonlinear function. Nonlinear neural network models which incorporate sigmoid function become popular in many time series applications due to their flexibility in handling time series data with nonlinearity. [14] investigated the coverage probabilities of neural network regression models and obtained long-term electricity demand prediction intervals for daily electricity demand in Victoria, Australia, and showed the superiority of neural network regression models over the dynamic regression models with seasonal ARIMA errors (DRSARIMA). In [15], the superior performance of the forecasts based on neural network regression models over the DRSARIMA and Prophet models has been demonstrated for Ontario, Canada, electricity demand data.

When obtaining one-step-ahead forecasts of the electricity demand, the models rely on the point forecasts of the temperature as well as the interval estimates. It is crucial to acknowledge that temperature prediction models are not completely accurate, and temperatures can fluctuate throughout the day. Additionally, in larger regions like Ontario, Canada, variations in local temperatures may be observed. This paper introduces a novel approach by treating temperature, a pivotal factor in electricity demand forecasting, as a fuzzy number to account for uncertainty. Furthermore, we propose a hierarchical modeling strategy to generate fuzzy electricity demand forecasts. First, the study derives fuzzy temperature forecasts based on various factors such as snowfall and air density. Then, these fuzzy temperature forecasts, combined with calendar characteristics, are utilized to obtain fuzzy electricity demand forecasts. More specifically, the proposed strategy can be summarized as a multi steps procedure:

Step 1: Obtain Fuzzy Forecasts of temperature.

Step 1a: Identify the most important covariate affecting temperature.

Step 1b: Fit different models and identify the best fit for temperature using forecast accuracy.

Step 1c: Model temperature as a nonlinear adaptive fuzzy number and obtain fuzzy forecasts of temperature.

Step 2: Obtain Fuzzy Forecasts of electricity demand.

Step 2a: Fit different models and identify the best fit for electricity using forecast accuracy.

Step 2b: Obtain fuzzy forecasts of electricity demand using the model selected in step 2a and fuzzy forecasts obtained in step 1c.

The remainder of the paper is organized as follows. In Section II, details of fuzzy forecasts using nonlinear adaptive fuzzy numbers and neural net dynamic regression models are provided. In Section III, experiments and results are described. Finally, concluding remarks are given in Section IV.

II. METHODOLOGY

The model introduced in this study aims to generate fuzzy forecasts for electricity demand by utilizing fuzzified temperature forecasts. In this section, we provide specifics of nonlinear

adaptive fuzzy numbers and the neural network autoregression model, identified as the optimal model for both temperature and electricity demand.

A. Regression Models with Correlated Errors

Regression models with correlated errors allow for time series dynamics where errors from a regression contain autocorrelation. Consider the model

$$y_t = \beta_0 + \beta_1 x_t + \eta_t \quad \text{with} \quad \eta_t = \phi \eta_{t-1} + \epsilon_t$$

where ϵ_t is a white noise series and β_0, β_1 and ϕ are known constants. η_t s are the errors from the regression model. It can be shown that the minimum mean square error h steps ahead forecast of observed series y_t can be written as

$$\hat{y}_{t+h} = \beta_0 + \beta_1(x_t + l) + \phi^l(y_t - \beta_0 - \beta_1 x_t).$$

That is the future forecasts of the y_t series depend on the residuals. Similarly, in dynamic regression models with SARIMA errors, we need to forecast the regression part and add the appropriate functions of the residuals to get the forecasts of y_t .

B. Fuzzy Forecasts Using Nonlinear Adaptive Fuzzy Numbers

Fuzzy numbers are used to model uncertainty and imprecision in real-world data, where boundaries or membership degrees are not precisely defined. Let R represent a set of all real numbers. A fuzzy number $A(x)$, $x \in R$ has the following form: $A(x) = g(x)$ if $x \in [x_1, x_2]$, $A(x) = 1$ if $x \in [x_2, x_3]$, $A(x) = h(x)$ if $x \in [x_3, x_4]$, and $A(x) = 0$ otherwise. Here functions g and h are both real functions. However, function g is a right continuous and increasing function, whereas function h is a left continuous and decreasing function. The parameters x_1, x_2, x_3, x_4 take real values satisfying $x_1 < x_2 < x_3 < x_4$. A discussion of fuzzy numbers can be found in Zimmermann [16].

The nonlinear adaptive asymmetric fuzzy number A can be denoted as $A = [x_1, x_2, x_3, x_4]_{m,n}$ and functions g and h are further defined as

$$g(x) = \left(\frac{x - x_1}{x_2 - x_1} \right)^m \quad \text{and} \quad h(x) = \left(\frac{x_4 - x}{x_4 - x_3} \right)^n,$$

where m and n are positive real numbers ($m, n > 0$) (Figure 1). When m and n are both equal to 1, the notation of the fuzzy number simplifies to $A = [x_1, x_2, x_3, x_4]$, and this is commonly known as a trapezoidal fuzzy number, which is a linear fuzzy number. The horizontal representations of fuzzy sets are the α -cuts. The α -cuts of the asymmetric adaptive nonlinear fuzzy numbers $A = (x_1, x_2, x_3, x_4)_{m,n}$ are given by $x_1 + \alpha^{\frac{1}{m}}(x_2 - x_1), x_4 - \alpha^{\frac{1}{n}}(x_4 - x_3)$ for all $\alpha \in [0, 1]$.

For this study, (x_1, x_2, x_3, x_4) are chosen as 5th, 25th, 75th, and 95th percentiles of innovation residuals from a forecasting model for daily temperature (x). Moreover, for simplicity, the study considers the case $m = n$. Therefore, with observed temperature data until time t , the h -step ahead fuzzy temperature forecasts (lower limit (LL) and upper limit(UL)) can be computed as

$$x_{Fuzz,LL} = \hat{x}_{t+h} + [x_1 + \alpha^{\frac{1}{m}}(x_2 - x_1)],$$

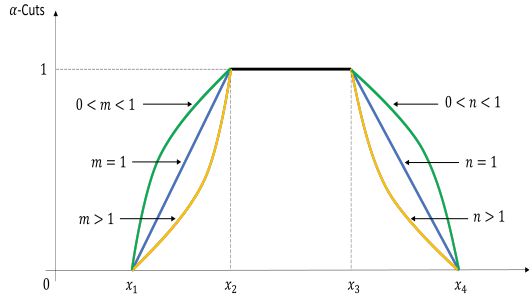


Fig. 1. Nonlinear membership function

and

$$x_{Fuzz,UL} = \hat{x}_{t+h} + [x_4 - \alpha^{\frac{1}{n}}(x_4 - x_3)],$$

where $h = 1, \dots, l$ and $\hat{x}_{t+h}$ are the daily temperature point forecasts.

It is observed that daily electricity demand exhibits a nonlinear relationship with temperature. Thus, we also obtain the square of fuzzy temperature forecasts, and the forecasts will be used to obtain fuzzy electricity demand forecasts. The square of a fuzzy number is a mathematical operation applied to fuzzy numbers, which are a part of fuzzy set theory. The concept of squaring a fuzzy number involves performing the square operation on each of the parameters that define the fuzzy number. A trapezoidal fuzzy number, denoted as $[x_1, x_2, x_3, x_4]$ is characterized by four parameters x_1, x_2, x_3 , and x_4 , where x_1 and x_4 represent the lower and upper bounds of the fuzzy number while x_2 and x_3 represent the left and right shoulders. Squaring a fuzzy number involves applying the square operation to each parameter.

$$\begin{aligned} A.A &= (x_1, x_2, x_3, x_4) \cdot (x_1, x_2, x_3, x_4) \\ &= (a, b, c, d) \end{aligned}$$

where

$$\begin{aligned} a &= \min(x_1x_1, x_1x_4, x_4x_1, x_4x_4), \\ b &= \min(x_2x_2, x_2x_3, x_3x_2, x_3x_3), \\ c &= \max(x_2x_2, x_2x_3, x_3x_2, x_3x_3), \\ d &= \max(x_1x_1, x_1x_4, x_4x_1, x_4x_4). \end{aligned}$$

Further details of a square of a fuzzy number and operations of fuzzy numbers are available in [17].

Once the fuzzy forecasts of temperature are obtained, they are used as inputs to obtain fuzzy forecasts of electricity demand (y) as

$$\hat{y}_{t+h, Fuzz} \sim f(x_{Fuzz}, x_{Fuzz}^2).$$

C. Neural Network Autoregression (NNAR)

The feed-forward neural network consists of an input variable(s), an output variable, and a hidden layer(s), which join the input variable(s) and output variable. The hidden layer(s) of the neural network is created with units called neurons. Neurons are capable of identifying new features from the data, and this process of learning is known as supervised learning.

With a sufficient number of neurons in the hidden layer(s), a neural network can approximate any continuous function, and the functions that translate inputs to desired output signals are known as transfer functions.

With time series data, lagged values of the time series can be used as inputs to a neural network, and it is called a neural network autoregression (NNAR) model. One of the popular NNARs specifically designed to handle time series data is *nnetar* proposed in [11]. It can be implemented in R using the *nnetar* function available in the *forecast* package, and it fits a single-layer neural network for the output variable considering its lag values. Also, it can be modified to consider some feature values [12], [13] when training the models. Thus, in such cases, it fits an NNAR(p, P, k)_m model, where p and P are the autoregressive (AR) orders of the non-seasonal and seasonal parts, k is the number of nodes in the hidden layer and m is the seasonal period. Unlike in the other feed-forward neural network models, in NNAR models, we can determine the number of neurons in the hidden layer by using $k = (p + P + 1)/2$.

The output variable (y_{t+h}) depends on the lagged values of the output variable ($y_{t+h-1}, y_{t+h-2}, \dots, y_{t+h-p}$) and as well as on the values of some features ($x_{1,t}, \dots, x_{L,t}$). However, if there are categorical features to be included in the model, they need to be encoded before feeding them as inputs in a neural network. It is important to remember that *nnetar* only fits a single-layer neural network, and all the neural networks constructed in this study have one hidden layer. A general representation of a neural network with p number of inputs, one hidden layer, and one output is given in Figure 2.

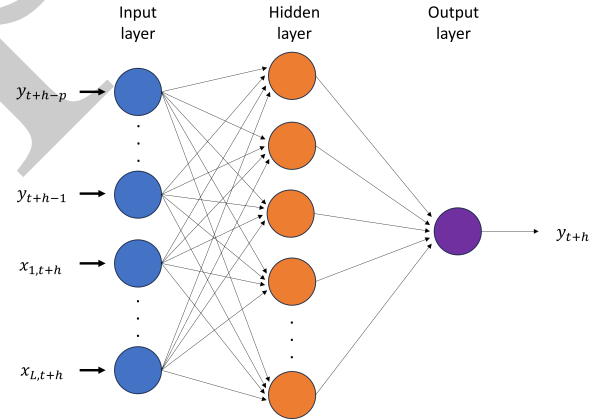


Fig. 2. Illustration of a neural network

In this study, *nnetar* is used to fit the NNAR model, where some features with the lagged values of the output variable are used as inputs of the neural network. In addition to NNAR, we have also used benchmark, prophet, ARIMA, DRSARIMA, and TSLM (time series linear model) models in this study. Detailed descriptions of these models are omitted here due to space constraints. However, references to various models discussed in the study have been provided in the introduction.

III. EXPERIMENTAL RESULTS

In this paper, we utilize a dataset containing daily electricity demand and meteorological information in Ontario, Canada, spanning from January 2018 to December 2019. The electricity demand (in Megawatts (MW)) data for the study period are obtained from Independent Electricity System Operator and meteorological feature values are obtained from Renewables.ninja. The primary objective is to generate one-step-ahead fuzzy forecasts for electricity demand using a hierarchical modeling approach, incorporating fuzzy one-step-ahead temperature forecasts.

Existing literature underscores the successful modeling of electricity demand by integrating calendar features and temperature. Our preliminary analysis indicates that additional meteorological factors, such as snowfall and air density, contribute to an enhanced model fit when employed to forecast temperature rather than electricity demand. Thus, in the first phase, the study incorporates meteorological influences including precipitation (mm/hour), snowfall (mm/hour), snow mass (kg/m²), air density (kg/m³), ground-level solar irradiance (W/m²), top-of-atmosphere solar irradiance (W/m²), and cloud cover fraction to generate fuzzy temperature. In the second phase, to generate fuzzy electricity demand forecasts, the models include the fuzzy temperature forecasts from the first phase and day type (weekday/weekend/holiday) information.

In order to check the multicollinearity issue, we have obtained the Variance Inflation Factor (VIF) of the meteorological features. VIF is a measure that quantifies how much the variance of an estimated regression coefficient increases if the features are correlated. High VIF values (typically > 10) may indicate problematic levels of multicollinearity. Table I provides the VIF values of the features, and Figure 3 visualizes the correlation among the features. VIF for Irradiance (surface) is greater than 10, and for Irradiance (atmosphere) is close to 10. The correlation matrix from Figure 3 indicates Irradiance (surface) and Irradiance (atmosphere), and Air Density, Irradiance (surface) and Cloud Cover, and Irradiance (surface) and Air Density are highly correlated. Thus, we omit Irradiance (surface) and Irradiance (atmosphere) from future models for temperature to avoid multicollinearity.

TABLE I
VARIANCE INFLATION FACTOR (VIF)

Precipitation	Irradiance (surface)	Irradiance (atmosphere)	Snowfall
1.89	10.64	8.48	1.73
Snow Mass	Cloud Cover	Air Density	
2.33	2.90	3.59	

Before obtaining temperature forecasts, it is crucial to determine the most suitable model for the available data. The study has explored thirteen different models, including benchmark models such as Mean, Naive, Seasonal Naive (SNAIVE), and Drift, with the corresponding test errors summarized in Table II. While simpler models like Mean and Naive lack

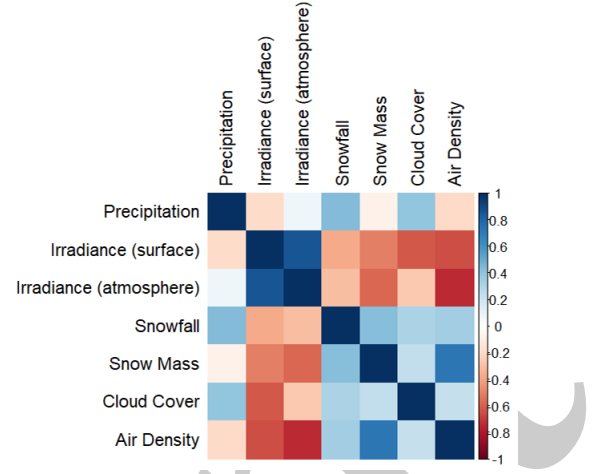


Fig. 3. Correlation Matrix

the capacity to incorporate additional features, more complex models such as Prophet, ARIMA, DRSARIMA, NNAR, and TSLM can accommodate regression-type time models for the given data. Two sets of models have been fitted for Prophet, NNAR, and TSLM. For instance, Prophet.1 involves fitting a Prophet model for temperature considering its lag values, while Prophet.2 fits a regression-type model for temperatures by integrating other meteorological features as

$$\text{Temperature} \sim f(\text{Precipitation}, \text{Snowfall}, \text{SnowMass}, \text{CloudCover}, \text{AirDensity}).$$

Similarly, NNAR.1 and NNAR.2, TSLM.1, and TSLM.2 are defined. Additionally, the Exponential Smoothing State Space (ETS) model is also used in this study. Table II presents the forecast accuracy for all temperature models.

TABLE II
FORECAST ACCURACY OF THIRTEEN MODELS FOR TEMPERATURE

Model	RMSE	MAE	MPE	MAPE
MEAN	11.05	9.71	193.47	581.53
NAIVE	13.58	10.90	95.51	113.97
SNAIVE	13.06	10.60	65.31	157.73
Drift	11.31	9.51	38.46	507.21
Prophet.1	3.34	2.55	-6.16	235.92
Prophet.2	1.23	0.98	-10.77	92.27
ARIMA	13.59	10.91	95.33	114.70
DRSARIMA	3.72	3.02	7.15	130.57
NNAR.1	12.43	10.34	213.52	1017.31
NNAR.2	0.58	0.40	-0.76	21.99
ETS	13.54	10.87	96.40	110.42
TSLM.1	11.05	9.71	193.47	581.53
TSLM.2	1.69	1.37	-18.20	106.10

RMSE - Root Mean Squared Error, MAE - Mean Absolute Error, MPE - Mean Percentage Error, MAPE - Mean Absolute Percentage Error

Based on the results in Table II, NNAR.2 provides the best model to obtain one-step-ahead fuzzy temperature forecasts and the model can be written as NNAR($p = 27, P = 1, k = 16$)₇ which contains 16 neurons in the hidden layer with the

past 27 observations of temperature ($y_{t-1}, \dots, y_{t-27}$) as inputs with meteorological features to obtain temperature forecasts. Moreover, the model suggests weekly seasonality ($m = 7$).

The point forecasts of temperature using NNAR.2 over the study period are visualized in Figure 4. It can be seen that forecasts are very close to the observed temperature during the study period. However, to capture the uncertainty associated with the point forecasts, we calculated the fuzzy temperature forecasts.

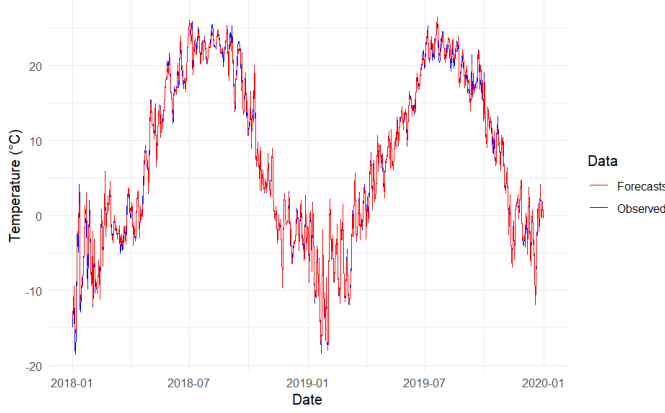


Fig. 4. Point forecasts of temperature from January 2018 to December 2019 using NNAR.2

One Step-ahead Fuzzy Forecasts of Temperature using NNAR.2 are given in Table III. The table provides fuzzy one-step-ahead forecasts (one day) of the temperature. However, this can be easily extended to obtain one-step-ahead fuzzy forecasts of temperature for several days or preferred horizon.

TABLE III
ONE STEP-AHEAD FUZZY FORECASTS OF TEMPERATURE USING NNAR.2

α	$n = 0.5$		$n = 1.0$		$n = 2.0$	
	LL	UL	LL	UL	LL	UL
0.0	-0.3652	1.9317	-0.3652	1.9317	-0.3652	1.9317
0.1	-0.3560	1.9203	-0.2727	1.8179	-0.0728	1.5720
0.2	-0.3282	1.8862	-0.1803	1.7042	0.0483	1.4230
0.3	-0.2820	1.8293	-0.0878	1.5904	0.1413	1.3087
0.4	-0.2173	1.7497	0.0047	1.4767	0.2196	1.2123
0.5	-0.1340	1.6473	0.0971	1.3629	0.2887	1.1274
0.6	-0.0323	1.5222	0.1896	1.2492	0.3511	1.0506
0.7	0.0879	1.3743	0.2821	1.1354	0.4084	0.9800
0.8	0.2266	1.2037	0.3745	1.0217	0.4619	0.9143
0.9	0.3838	1.0103	0.4670	0.9079	0.5120	0.8526
1.0	0.5595	0.7942	0.5595	0.7942	0.5595	0.7942

LL - Lower limit, UL - Upper limit

After obtaining the fuzzy temperature forecasts, the subsequent step involves acquiring fuzzy forecasts for electricity demand. As outlined earlier, our approach involves modeling electricity demand based on fuzzy temperature forecasts, square of fuzzy temperature forecasts, and day type. Despite NNAR providing the best-fit temperature, it does not guarantee the same optimal fit for electricity demand. Thus, we fit different models for electricity demand and choose the best model to obtain fuzzy electricity demand forecasts. Moreover,

considering the seasonal variations in Ontario's temperature, it becomes imperative to explore the relationship between temperature and electricity demand before choosing the best-fit model for electricity demand.

Figure 5 visualizes the changes in electricity demand with temperature for Ontario throughout the study period. The average daily temperature in Ontario fluctuates within the range of -20°C to 30°C , with electricity demand peaking during periods of high or very low daily temperatures. It is reasonable to infer that during colder periods, heating contributes to increased electricity demand, while elevated temperatures lead to higher demand due to air conditioning. Additionally, Figure 5 incorporates day type (weekday, weekend, or holiday), revealing that electricity demand is notably higher on weekdays compared to weekends. Furthermore, the daily electricity demand exhibits a nonlinear relationship with temperature.

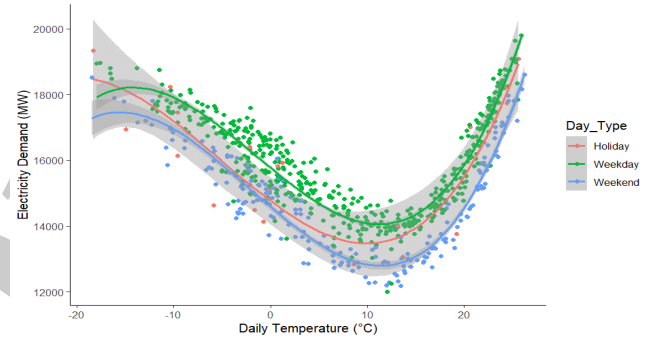


Fig. 5. Electricity demand versus Temperature from January 2018 to December 2019

Various options exist for modeling a nonlinear relationship. Nevertheless, this study focuses on exploring four widely used models (TSLM, DRSARIMA, NNAR, and Prophet) in the time series context to identify the optimal model for electricity demand considering temperature and day type (DayType). For this purpose, we have allocated 80% of the available data (from 2018-01-01 to 2019-08-07) for training the models. All four models have been applied, and the corresponding forecast accuracy based on test data (from 2019-08-08 to 2019-12-31) for each model is presented in Table IV. Notably, among these four models, NNAR demonstrates the highest forecast accuracy in terms of RMSE, MAE, and MAPE. The optimal NNAR model is configured with 14 neurons in the hidden layer, utilizing the preceding 23 temperature observations ($y_{t-1}, \dots, y_{t-23}$) as inputs to generate temperature forecasts and can be written as $\text{NNAR}(p = 23, P = 1, k = 14)_7$. Additionally, the model also indicates a weekly seasonality ($m = 7$). Given the nonlinear nature of the relationship between electricity demand and temperature, it is also reasonable to anticipate that NNAR holds the potential to provide the most accurate forecasts for electricity demand, considering how well a neural network can approximate a nonlinear relationship.

Finally, the one-step-ahead fuzzy forecasts for electricity demand are calculated using the regression-type NNAR model

TABLE IV
FORECAST ACCURACY OF FIVE REGRESSION MODELS FOR ELECTRICITY DEMAND

Model	RMSE	MAE	MPE	MAPE
TSLM	877.95	777.70	-1.01	5.21
DRSARIMA	1443.74	1246.76	0.49	8.17
NNAR	734.44	561.19	-0.08	3.70
Prophet	2564.53	1967.14	12.19	12.61

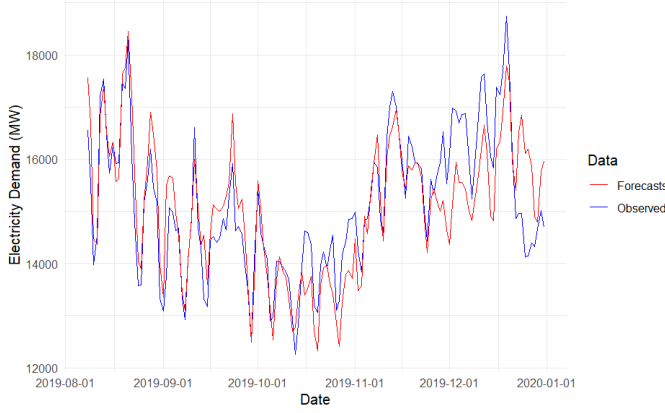


Fig. 6. Electricity demand forecast from August 2018 to December 2019 using NNAR

as

$$\text{Electricity Demand} \sim f(\text{Temperature}_{Fuzz}, \text{Temperature}_{Fuzz}^2, \text{DayType})$$

Table V presents one-step-ahead fuzzy forecasts for electricity demand on the final day of the study period. Here, the table provides fuzzy electricity demand forecasts across α values ranging from 0 to 1, with a step size of 0.1, and for n values of 0.5, 1.0, and 2.0. Consequently, when considering α and n at 0.5 and 1.0, respectively, the anticipated range for electricity demand falls within (15387.72, 15474.72).

TABLE V
ONE STEP-AHEAD FUZZY FORECASTS OF ELECTRICITY DEMAND USING NNAR

α	$n = 0.5$		$n = 1.0$		$n = 2.0$	
	LL	UL	LL	UL	LL	UL
0.0	15345.11	15503.15	15346.36	15507.84	15343.61	15503.86
0.1	15347.35	15507.72	15353.68	15499.61	15372.11	15487.91
0.2	15346.68	15502.14	15367.33	15491.32	15387.14	15481.67
0.3	15351.89	15504.35	15372.24	15489.40	15390.12	15468.45
0.4	15360.91	15492.88	15381.96	15478.20	15402.60	15469.32
0.5	15364.75	15493.34	15387.72	15474.72	15404.76	15458.80
0.6	15377.21	15484.13	15400.24	15464.93	15414.05	15458.10
0.7	15392.70	15476.32	15408.66	15458.45	15415.49	15448.66
0.8	15400.15	15462.95	15415.93	15456.54	15425.89	15445.49
0.9	15415.15	15450.08	15427.27	15443.72	15429.12	15446.36
1.0	15432.24	15438.54	15430.71	15439.32	15438.82	15439.05

IV. CONCLUSIONS

Recent research indicates that enhancing electricity demand forecasts involves incorporating essential features like calendar

characteristics and meteorological effects into statistical and machine-learning models. The superiority of fuzzy forecasts obtained by incorporating randomness and fuzziness in option pricing and fuzzy algo trading has been demonstrated in the literature.

Temperature stands out as a pivotal factor in electricity demand forecasting, and existing literature often relies on point forecasts or specific temperature values to generate electricity demand forecasts. Nonetheless, temperature is a highly volatile feature, and using point forecasts for temperature may lead to inaccuracies in electricity demand forecasts. In this paper, the driving idea, unlike the existing work, is to demonstrate the use of fuzzy temperature forecasts to obtain fuzzy electricity demand forecasts while accommodating the uncertainty in temperature forecasts. Thirteen different models, including neural network regression models and Facebook industrial Prophet models, are also fitted, and it is shown that the neural network regression models outperform other models in terms of forecast accuracies.

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